

Python Data Visualization Project Using Seaborn

Student Task Sheet

Objective

The objective of this project is to learn how to use **Python, Pandas, Matplotlib, and Seaborn** to analyze real datasets and represent the findings using proper data visualizations.

Students have to choose **one dataset**, write Python code, create suitable graphs, and explain the analysis.

Tools Required

Students should use the following Python libraries:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Project Task

Choose **any one dataset** from the list below and complete the given tasks.

Dataset 1: Titanic Survival Analysis

Dataset

```
df = sns.load_dataset("titanic")
```

Task Questions

1. Display the first 5 rows of the dataset.
2. Find the number of rows and columns.
3. Check missing values in the dataset.
4. Find how many passengers survived and how many did not.
5. Compare survival based on gender.
6. Compare survival based on passenger class.
7. Show the age distribution of passengers.
8. Analyze whether fare and age have any relation with survival.

Suggested Graphs

Analysis	Seaborn Plot
Survival count	countplot
Gender vs survival	countplot
Class vs survival	barplot
Age distribution	histplot
Age vs fare	scatterplot

Final Analysis Questions

1. Which gender had a higher survival rate?
2. Which passenger class had the highest survival rate?
3. What was the common age group of passengers?
4. Did higher fare passengers have better survival chances?

Dataset 2: Tips Dataset Analysis

Dataset

```
df = sns.load_dataset("tips")
```

Task Questions

1. Display the first 5 rows.
2. Find the average total bill.
3. Find the average tip amount.
4. Compare total bill on different days.
5. Analyze whether higher bills lead to higher tips.
6. Compare tips given during lunch and dinner.
7. Compare tips based on smoker and non-smoker customers.
8. Find which day has the highest average tip.

Suggested Graphs

Analysis	Seaborn Plot
Total bill vs tip	scatterplot
Day-wise bill distribution	boxplot
Tips by gender	violinplot
Time vs tip	barplot
Smoker vs tip	boxplot

Final Analysis Questions

1. Do customers with higher bills give higher tips?
2. Which day has the highest total bill?
3. Do lunch or dinner customers give more tips?
4. Do smokers and non-smokers show different tipping behavior?

Dataset 3: Penguin Species Analysis

Dataset

```
df = sns.load_dataset("penguins")
```

Task Questions

1. Display the dataset.
2. Find the unique penguin species.
3. Check missing values.
4. Compare body mass of different penguin species.
5. Compare flipper length of different species.
6. Analyze the relation between flipper length and body mass.
7. Compare male and female penguins based on body mass.
8. Use pairplot to compare all numerical features.

Suggested Graphs

Analysis	Seaborn Plot
Species count	countplot
Body mass by species	boxplot
Flipper length vs body mass	scatterplot
Gender-wise body mass	violinplot
Feature comparison	pairplot

Final Analysis Questions

1. Which penguin species has the highest body mass?
2. Is there a relation between flipper length and body mass?
3. Which features help in separating penguin species?
4. Are male penguins heavier than female penguins?

Dataset 4: Iris Flower Dataset

Dataset

```
df = sns.load_dataset("iris")
```

Task Questions

1. Display the first 5 rows.
2. Find all flower species in the dataset.
3. Find the average sepal length and petal length.
4. Compare petal length of different species.
5. Compare sepal width of different species.
6. Create a correlation heatmap.
7. Use pairplot to visualize species separation.
8. Identify which feature best separates the species.

Suggested Graphs

Analysis	Seaborn Plot
Species count	countplot
Petal length comparison	boxplot
Sepal width comparison	violinplot
Correlation	heatmap
Overall comparison	pairplot

Final Analysis Questions

1. Which species has the longest petals?
2. Which features are strongly correlated?
3. Which species is easiest to identify visually?
4. Which graph gives the best overall comparison?

Dataset 5: Student Performance Analysis

Dataset Description

Students can use the **Students Performance in Exams** dataset from Kaggle.

The dataset usually contains the following columns:

gender
race/ethnicity
parental level of education
lunch
test preparation course
math score
reading score
writing score

Task Questions

1. Load the CSV file using Pandas.
2. Display the first 10 rows.
3. Check missing values.
4. Find average marks in math, reading, and writing.
5. Compare marks based on gender.
6. Analyze the effect of test preparation course.
7. Compare marks based on parental education.
8. Create a new column named `average_score`.
9. Find which factor affects student performance the most.
10. Create a heatmap showing correlation among scores.

Suggested Graphs

Analysis	Seaborn Plot
Gender vs marks	boxplot
Test preparation vs marks	barplot
Parental education vs marks	barplot
Score distribution	histplot
Score correlation	heatmap

Final Analysis Questions

1. Does test preparation improve marks?
2. Which subject has the highest average score?
3. Is there a strong relation between reading and writing scores?
4. Does parental education affect student performance?
5. What is your final conclusion from the dataset?

Common Instructions for All Students

Students must complete the following sections in their project.

1. Introduction

Write 5–6 lines about:

- Dataset name
- Source of dataset
- Number of rows and columns
- Purpose of analysis

2. Import Libraries

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

3. Load Dataset

Example for Seaborn dataset:

```
df = sns.load_dataset("titanic")
```

Example for CSV dataset:

```
df = pd.read_csv("student_performance.csv")
```

4. Basic Data Exploration

Students must write code for:

```
df.head()
df.shape
df.info()
df.describe()
df.isnull().sum()
```

5. Data Visualization

Students must create at least **5 different graphs** using Seaborn.

Compulsory graph types:

1. Countplot
2. Barplot
3. Boxplot
4. Histogram
5. Heatmap or Scatterplot

6. Observation After Each Graph

After every graph, students must write 2–3 lines explaining what they understood.

Example:

The graph shows that female passengers had a higher survival count than male passengers. This means gender played an important role in survival during the Titanic disaster.

7. Final Conclusion

Students must write a final conclusion of 8–10 lines based on the complete analysis.

Sample Project Structure

1. Project Title
2. Introduction
3. Dataset Description
4. Importing Libraries
5. Loading the Dataset
6. Data Cleaning
7. Exploratory Data Analysis
8. Data Visualization
9. Observations
10. Final Conclusion
11. Python Code

Submission Instructions

Students should submit:

1. Python notebook file: `.ipynb`
2. Dataset file if used externally: `.csv`
3. Final report: PDF or Word file
4. Minimum 5 graphs
5. Explanation after each graph
6. Final conclusion

Evaluation Criteria

Criteria	Marks
Dataset loading and exploration	10
Data cleaning	10
Correct use of Seaborn graphs	20
Quality of visualizations	20
Observations after each graph	20
Final conclusion	10
Code readability	10
Total	100

Suggested Project Title

Exploratory Data Analysis and Visualization Using Seaborn on a Real Dataset